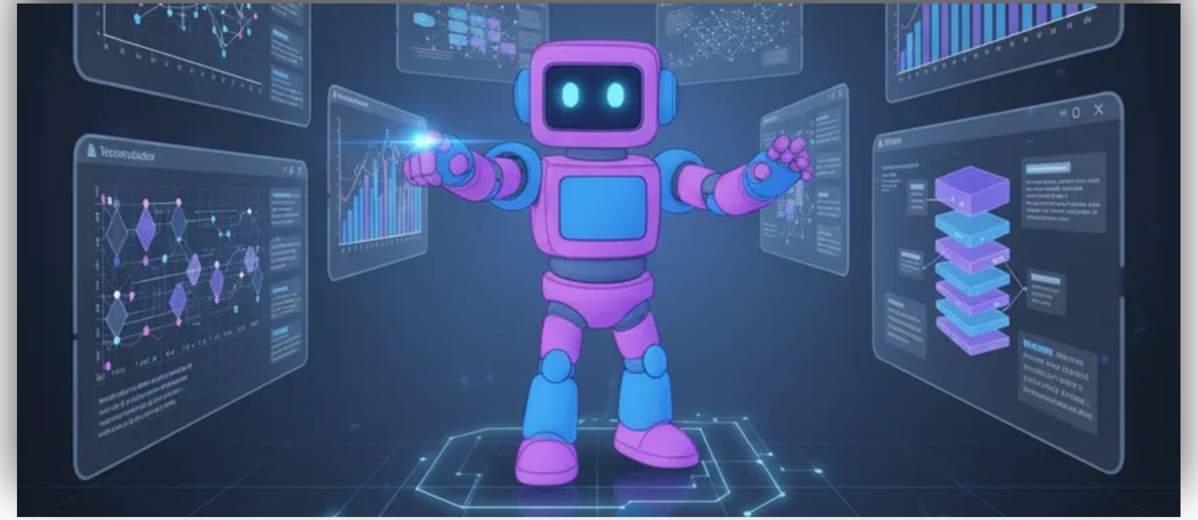
WEEK 2 – LECTURE 1

Artificial Intelligence Foundations

Formulating an AI Problem



State Space • Problem Characteristics • Search Space • Solution Space

Quick Recap: The AI Lifecycle

Today's lecture focuses entirely on the very first step.

Problem → Data → Model → Deployment

Before an AI can learn from data, we must precisely define what the problem is.

Motivation: Where to Begin?

Google Maps
Shortest Route

Solving a
Maze

Robot Navigating
a Room

How does an AI know where to begin solving these problems?

Formulating an AI Problem

AI must understand four core things before solving anything:

Initial Situation

↓

Available Actions

↓

Goal State

Real-World Analogy: Navigation

Current Location

↓

Possible Roads

↓

Destination

AI views every problem in this structured manner.

Understanding State Space

State Space

The complete collection of all possible states (situations) a problem can have.

Tic-Tac-Toe

All possible board configurations

Maze

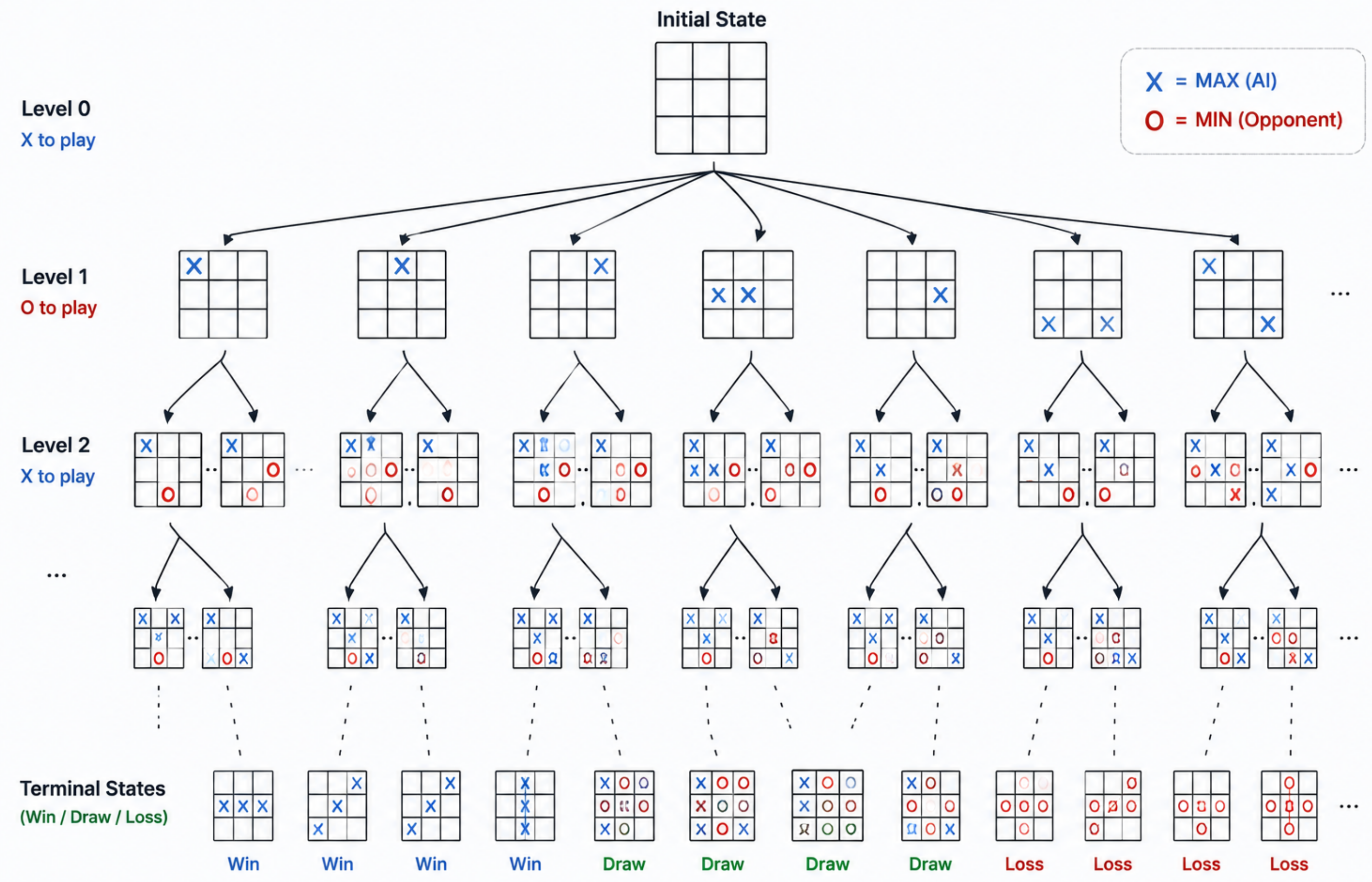
Every valid square you can stand on

Robot

Every valid coordinate in the room

Visualizing State Space

- Starts at an initial node.
- Every move branches out.
- Each node is a new situation.
- Continues until goal is found.



Interactive Activity

"Imagine solving a Rubik's Cube.

What would a single 'state' represent?"

Initial State vs. Goal State

AI always searches for a path from Start to Goal.

Maze

Start: Entrance → Goal: Exit

Chess

Start: Board Setup → Goal: Checkmate

Sudoku

Start: Empty Grid → Goal: Filled correctly

Robot

Start: Dock → Goal: Package delivered

Core Components of an AI Problem

Initial State + Actions + Goal State + Path Cost

Problem Characteristics

Not all AI problems are alike. Environments vary based on:

Fully vs Partially Observable

Can the AI see everything? (Chess vs Poker)

Deterministic vs Uncertain

Are outcomes guaranteed? (Puzzle vs Driving)

Static vs Dynamic

Does the world change while thinking?

Single vs Multi-Agent

Are there other entities? (Maze vs Chess)

Discrete vs Continuous

Fixed steps vs fluid motion?

Discussion: Which is the hardest for AI?

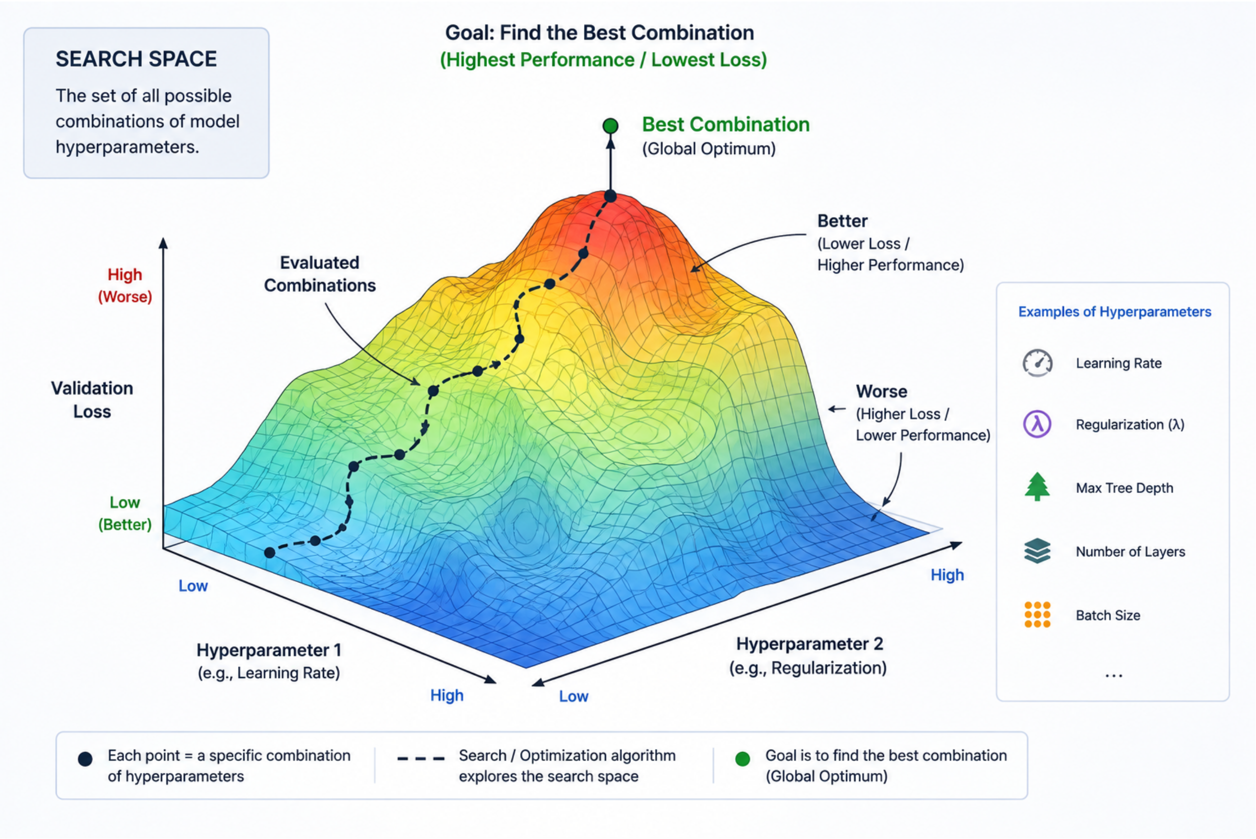
1. Chess

2. Self-Driving Car

3. Weather Prediction

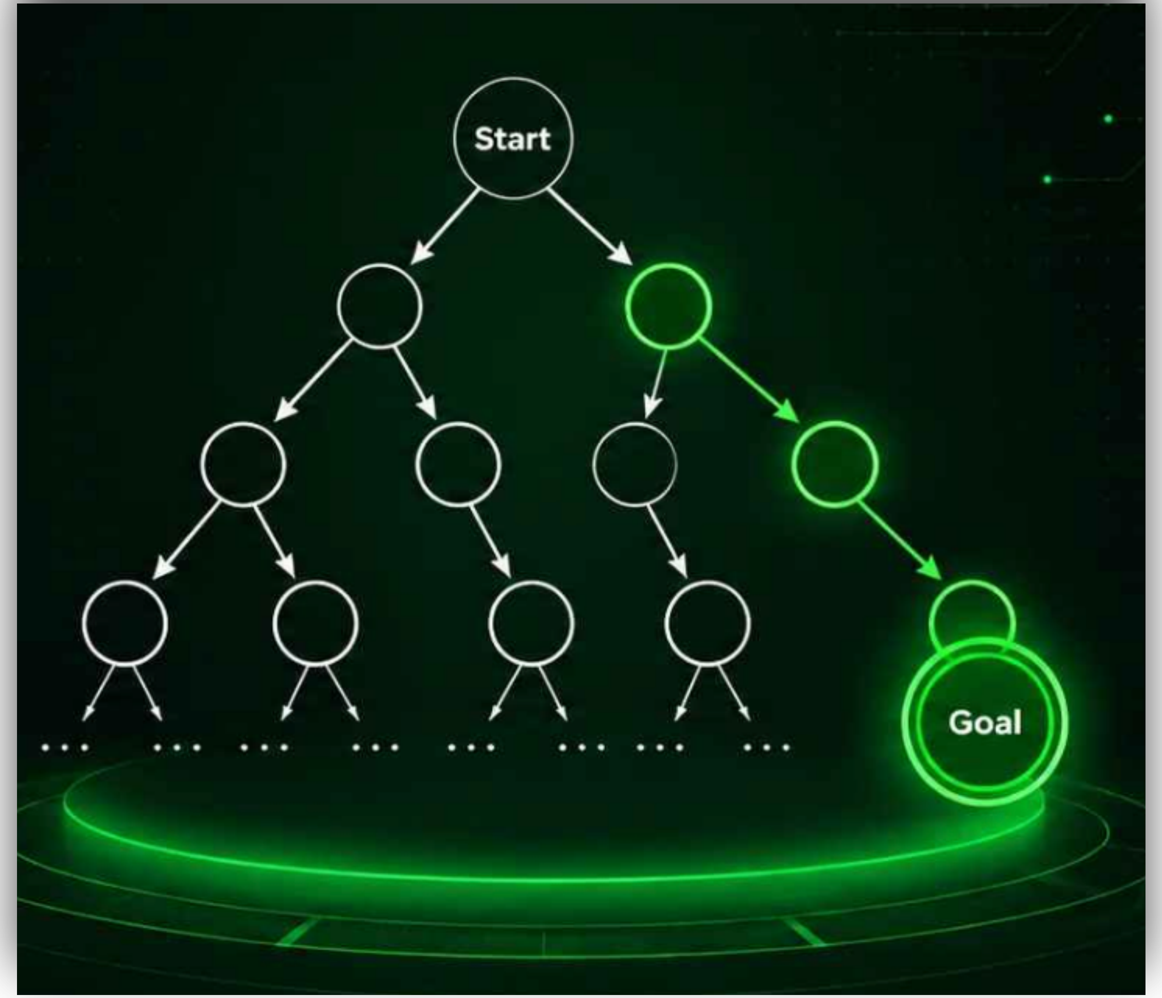
Search Space

- Every possible path the AI could explore.
- Includes dead ends.
- Dictates how long the AI takes to compute.



Solution Space

- Only the valid solutions.
- A subset of the larger search space.
- What we actually care about finding.



Search Space vs. Solution Space

Feature	Search Space	Solution Space
Size	Massive	Small
Failed Paths?	Yes	No
Successful Paths?	Yes	Yes

Case Study: Google Maps Route Planning

Thousands of possible roads

↓

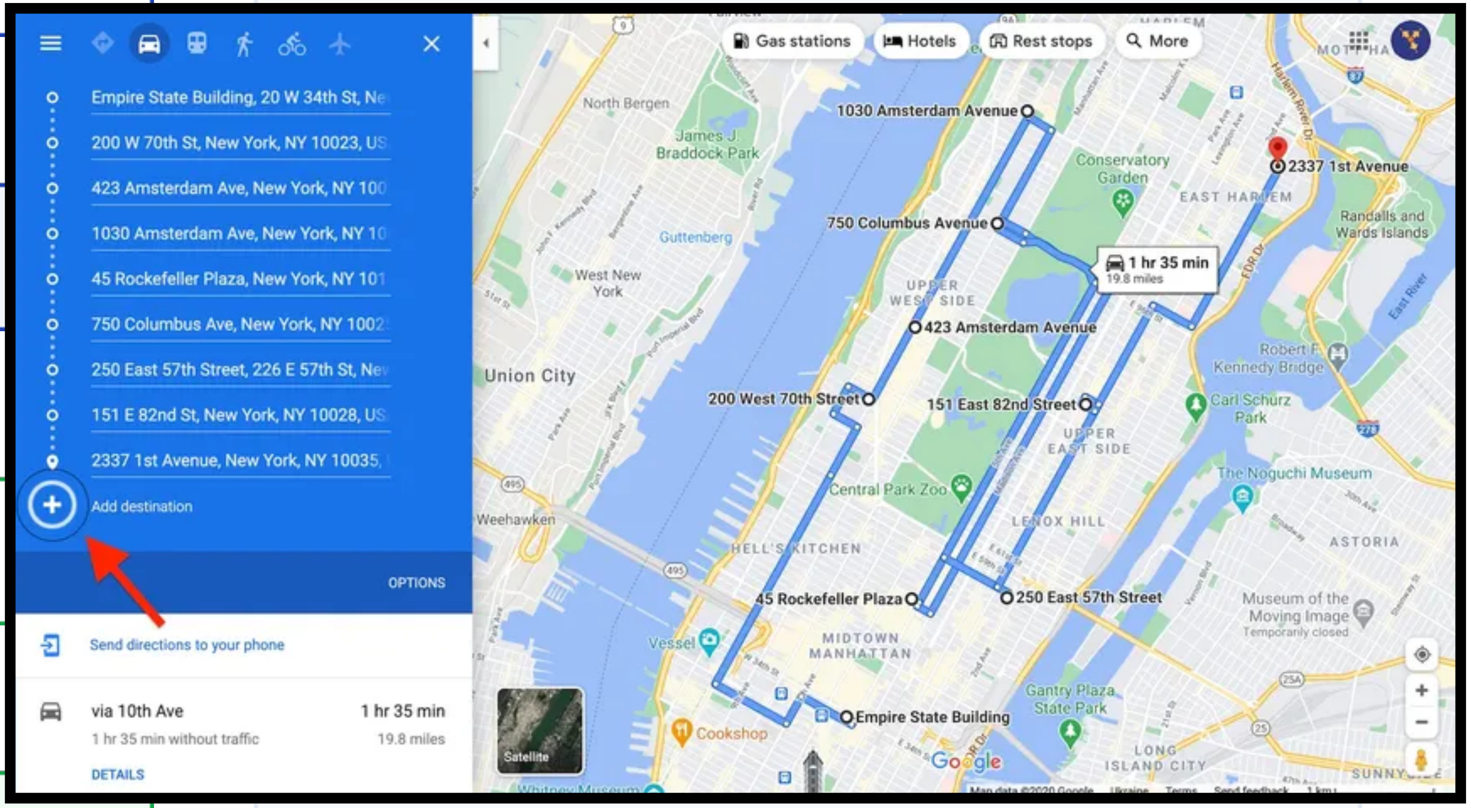
Search Space

↓

Shortest Valid Route

↓

Solution Space



Another Example: Sudoku

SUDOKU

2			3		6			
6		5	9			4		8
						5		2
4		9		6	3			
			8			7		1
		1		4			9	
1		6	2	7				
	2					8		4
		4		1	8			7

ANSWER

2	4	8	3	5	6	1	7	9
6	1	5	9	2	7	4	3	8
7	9	3	4	8	1	5	6	2
4	7	9	1	6	3	2	8	5
3	6	2	8	9	5	7	4	1
8	5	1	7	4	2	3	9	6
1	8	6	2	7	4	9	5	3
5	2	7	6	3	9	8	1	4
9	3	4	5	1	8	6	2	7

- **State:** Grid with numbers filled.
- **Search Space:** Millions of incorrect number combinations.
- **Solution Space:** The ONE correct completed puzzle.

Board Activity

Draw

- State-space tree
- Search space boundaries
- Solution path

Common Misconceptions

❌ **"Every search path is a solution."**

Correction: Most paths are dead ends.

❌ **"Bigger search spaces mean better AI."**

Correction: Bigger spaces are harder/slower to compute.

❌ **"State and search space are exactly the same."**

Correction: State space is possible situations; Search space is possible paths.

Key Takeaways

Problems must be formulated first.	States represent specific situations.	State space = all possible states.	Search space = exploration paths.	Solution space = valid solutions.

Quick Quiz

Which represents ALL possible exploration paths?

- A) Goal State
- B) Search Space
- C) Solution Space
- D) Initial State

Quick Quiz

Which represents ALL possible exploration paths?

- A) Goal State
- **B) Search Space**
- C) Solution Space
- D) Initial State

Preview: Next Lecture

Performance Measures & Problem-Solving Approaches

Problem → Search Algorithm → Optimal Solution

Image Sources

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